

THE PAS, MB, Canada

WMO: 718670

Lat: 53.9747N

Lon: 101.0889W

Elev: 270

StdP: 98.12

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-31.5	-38.7	0.1	-33.7	-36.4	0.1	-31.3	10.5	-15.5	9.6	-17.1	2.5	280	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	28.3	19.0	26.6	18.0	25.0	17.2	20.5	25.8	19.4	24.6	18.3	23.2	4.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.9	23.1	17.5	12.9	21.7	16.4	12.1	20.7	60.4	25.8	56.7	24.8	53.1	23.2	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.2	8.1	-37.5	32.1	2.6	1.7	-39.3	33.3	-40.9	34.4	-42.3	35.3	-44.2	36.6
			-37.6	23.0	2.6	1.6	-39.4	24.2	-41.0	25.1	-42.4	26.0	-44.3	27.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-18.8	-16.2	-8.9	0.8	8.5	15.3	18.6	17.2	11.5	3.5	-6.8	-15.6	(6)
(7)		DBStd	14.42	8.09	7.42	7.86	6.35	5.03	3.92	3.05	3.36	4.35	4.48	6.55	8.02	(7)
(8)		HDD10.0	4124	893	734	586	278	90	5	0	1	33	207	504	793	(8)
(9)		HDD18.3	6470	1151	968	844	524	307	105	34	63	207	461	754	1051	(9)
(10)		CDD10.0	782	0	0	0	3	42	164	266	222	79	5	0	0	(10)
(11)		CDD18.3	87	0	0	0	0	1	14	42	26	3	0	0	0	(11)
(12)		CDH23.3	743	0	0	0	0	26	142	333	212	31	0	0	0	(12)
(13)		CDH26.7	148	0	0	0	0	4	29	65	46	4	0	0	0	(13)
(14)	Wind	WSAvg	3.7	3.6	3.5	3.7	3.7	3.7	3.5	3.5	3.7	4.0	4.2	4.0	3.6	(14)
(15)	Precipitation	PrecAvg	469	20	15	20	25	39	68	77	64	59	38	25	21	(15)
(16)		PrecMax	606	61	64	67	97	100	192	211	147	129	120	61	52	(16)
(17)		PrecMin	319	4	1	1	1	8	20	13	7	5	1	5	1	(17)
(18)		PrecStd	75	12	10	13	18	22	38	41	35	31	25	14	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	3.7	9.6	20.0	26.2	29.8	30.6	30.5	26.7	18.7	8.8	2.7	(19)
(20)			MCWB	1.4	1.1	4.5	10.6	14.1	19.2	20.7	20.3	17.6	11.8	4.8	0.6	(20)
(21)		2%	DB	-0.9	0.8	6.7	16.3	22.9	26.9	28.4	27.7	23.6	15.0	5.6	-0.6	(21)
(22)			MCWB	-2.0	-0.5	2.8	8.8	13.0	17.4	19.3	18.9	16.3	10.0	3.0	-1.9	(22)
(23)		5%	DB	-4.0	-2.1	4.3	12.8	20.4	24.7	26.8	25.7	21.1	12.4	3.1	-3.0	(23)
(24)			MCWB	-4.8	-3.2	1.5	6.8	11.7	16.3	18.4	18.2	14.8	8.6	1.5	-3.8	(24)
(25)		10%	DB	-6.5	-4.9	2.0	10.0	17.7	22.7	25.1	23.7	18.6	10.0	1.4	-4.8	(25)
(26)			MCWB	-7.1	-5.8	0.0	5.0	10.5	15.2	17.6	17.1	13.8	7.3	0.2	-5.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.5	4.9	11.3	16.2	21.0	22.4	21.8	19.1	13.0	5.4	0.8	(27)
(28)			MCDWB	2.9	3.2	8.9	18.4	23.9	25.3	27.6	27.9	24.9	17.3	7.8	2.7	(28)
(29)		2%	WB	-2.0	-0.6	3.1	9.2	14.0	19.0	20.8	20.2	17.1	10.6	3.2	-1.8	(29)
(30)			MCDWB	-0.9	0.8	6.2	15.7	20.3	24.8	25.9	25.8	22.1	14.2	5.1	-0.7	(30)
(31)		5%	WB	-4.8	-3.2	1.7	7.2	12.5	17.6	19.6	19.1	15.6	8.9	1.6	-3.8	(31)
(32)			MCDWB	-4.0	-2.1	4.1	12.4	18.7	22.7	24.7	24.1	20.1	11.9	3.1	-3.1	(32)
(33)		10%	WB	-7.1	-5.8	0.2	5.2	11.1	16.2	18.6	17.9	14.1	7.4	0.1	-5.4	(33)
(34)			MCDWB	-6.5	-4.9	2.0	9.5	16.9	21.0	23.6	22.3	18.0	9.7	1.3	-4.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	10.1	11.2	11.0	11.4	10.4	10.1	9.8	9.3	7.0	6.5	7.5	(35)
(36)			MCDBR	9.9	10.3	11.4	14.6	16.0	13.4	12.8	13.0	13.4	11.5	8.0	8.6	(36)
(37)			MCWBR	9.1	8.8	8.2	8.9	8.1	6.1	5.8	6.3	7.3	7.1	6.0	7.7	(37)
(38)		5% WB	MCDBR	9.8	9.9	10.8	14.1	14.1	11.4	11.3	11.4	12.3	10.6	7.5	8.3	(38)
(39)			MCWBR	9.0	8.5	8.0	8.7	7.8	5.9	5.8	6.1	7.2	6.8	5.8	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.235	0.244	0.272	0.337	0.348	0.378	0.389	0.363	0.321	0.289	0.246	0.221	(40)	
(41)		taud	2.306	2.334	2.360	2.372	2.424	2.361	2.380	2.448	2.553	2.576	2.441	2.313	(41)	
(42)		Ebn at Noon	760	879	921	893	900	870	853	857	860	821	752	709	(42)	
(43)		Edh at Noon	55	75	92	106	107	115	111	98	78	61	48	44	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.94	2.08	3.59	4.96	5.50	5.58	5.68	4.66	3.23	1.78	0.83	0.62	(44)	
(45)		RadStd	0.09	0.14	0.29	0.39	0.54	0.42	0.36	0.36	0.35	0.24	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.49	N/A	-0.38	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	+1.62	N/A	-0.36	-0.41	N/A	N/A	N/A	N/A

Nomenclature: See separate page